

Assignment Questions on Turing Machines, Decidability, Cook–Levin, and Rice’s Theorem

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1. Design a Turing machine that decides the language

$$L = \{ 0^n 1^n \mid n \geq 1 \}.$$

Describe the high-level algorithm and justify why the machine halts for all inputs.

2. Given a Turing machine with some transition table (provided separately), determine whether the machine *accepts*, *rejects*, or *loops* on input 001. Show the sequence of configurations until the outcome becomes clear.

3. Construct a Turing machine that *recognizes but does not decide* the language

$$L = \{ w \in \{0, 1\}^* \mid w \text{ contains more 1's than 0's} \}.$$

Explain why the machine may loop on certain inputs.

4. Consider the language

$$\text{HALT}_\varepsilon = \{ \langle M \rangle \mid M \text{ halts on the empty string} \}.$$

Is this language decidable or undecidable? Provide a brief proof sketch.

5. Show that the language

$$L = \{ \langle M \rangle \mid M \text{ accepts at least one string of length } \leq 3 \}$$

is decidable. Describe a decision procedure for it.

6. Prove that the language

$$\text{EQ}_{\text{CFG}} = \{ \langle G_1, G_2 \rangle \mid L(G_1) = L(G_2) \}$$

is undecidable. Identify the known undecidable problem used in your reduction.

7. State the Cook–Levin theorem and briefly explain how configurations of a nondeterministic Turing machine are encoded as clauses in the Boolean formula produced in the reduction to SAT.
8. Suppose a nondeterministic Turing machine runs in $O(n^2)$ time. Explain (asymptotically) how many variables and clauses appear in the SAT instance constructed by the Cook–Levin reduction. Justify your bounds.
9. Use Rice’s theorem to show that the language

$$L = \{ \langle M \rangle \mid L(M) \text{ is finite} \}$$

is undecidable. Identify the semantic property involved and explain why the theorem applies.

10. Consider the language

$$P = \{ \langle M \rangle \mid M \text{ accepts some string containing the substring } 111 \}.$$

Determine whether P is decidable. Use Rice’s theorem when applicable, or justify carefully why it does or does not apply.